SOFTWARE ENGINEERING MINI-PROJECT

CHAT APPLICATION

Real-Time Chat Application Using WebSockets

SYSTEM VALIDATION REPORT

PROJECT DETAILS

Course: Software Engineering

Semester: 5th Semester

Section: K

Branch: Computer Science and Engineering (CSE)

University: PES University

Academic Year: 2026–2027

TEAM MEMBERS

Sl. No.	SRN	Name
1	PES2UG24CS625	Amoghaa R
2	PES2UG24CS664	Vaibhav R
3	PES2UG25CS817	P S Harshavardhan
4	PES2UG24CS648	Pramath Ganesh Hegde

SYSTEM VALIDATION REPORT

Real-Time Chat Application Using Socket Programming

1. Introduction

This report presents the system validation activities performed on the Real-Time Chat Application. The objective was to evaluate the application's functionality, performance, reliability, and responsive user interface against the requirements specified in the Software Requirements Specification (SRS) and Software Test Plan (STP).

2. Objectives

The main objectives of system validation were:

- Verify real-time message communication between users.
- Evaluate application performance under concurrent user load.
- Validate message delivery latency against the specified requirement.
- Verify automatic reconnection and message recovery.
- Evaluate responsive UI behaviour across different screen sizes.
- Assess whether the implemented application satisfies the tested functional and non-functional requirements.

3. Testing Environment

Component	Technology
Operating System	Windows
Programming Language	Python 3.14.2
Backend Framework	Flask and Flask-SocketIO
Communication	WebSocket / Socket.IO
Database	SQLite
Backend Testing	Pytest
Performance Testing	Python performance testing scripts
Frontend Testing	Node.js
Coverage Tool	pytest-cov

Browser Verification	Browser Developer Tools

4. Functional Validation

Functional validation covered the major application modules, including authentication, messaging, conversation access control, group communication, file sharing, presence notifications, and administrative operations.

The automated backend test suite was executed using the following command:

python -m pytest tests -v --junitxml=tests/results/unit_test_results.xml

The execution produced the following results:

Metric	Result
Total test cases	141
Passed	141
Failed	0
Skipped	0
Execution time	49.71 seconds

The results demonstrate that all executed backend test cases passed successfully.

5. Performance Validation

Performance validation was conducted to evaluate the application's ability to handle concurrent users while maintaining acceptable message delivery latency.

The specified requirement was to achieve a p95 message delivery latency of 200 milliseconds or less with 100 concurrent users.

An initial performance test identified latency issues. The team subsequently implemented performance optimizations under Issue #14, and the changes were merged into the main development branch.

The successful performance test result recorded on 10 November 2026 showed the following:

Metric	Result
Total concurrent users	100
Connected clients	100

Expected messages	250
Received messages	250
Lost messages	0
Message delivery rate	100%
Average latency	49.823 ms
Median latency	51.969 ms
p95 latency	70.029 ms
p99 latency	76.268 ms
Maximum latency	77.680 ms
Required p95 latency	≤ 200 ms
Test status	PASS

The application successfully connected all 100 simulated users and delivered all expected messages without message loss. The recorded p95 latency of 70.029 milliseconds was within the specified limit of 200 milliseconds.

Therefore, the performance requirement was satisfied in the recorded test execution.

6. Reliability and Reconnection Validation

Reliability testing was performed to evaluate the application's ability to recover from connection interruptions and restore communication.

The following results were obtained:

Metric	**Result**
Number of clients	10
Reconnection rounds	3
Total reconnection samples	30
Maximum reconnection time	2360 ms
p95 reconnection time	2155 ms
Mean reconnection time	996 ms

Clients that failed to reconnect	0
Messages lost	0
Observed availability	100%
Observation duration	35.4 seconds

All tested clients successfully reconnected, and no messages were lost during the recorded test.

The maximum reconnection time of 2360 milliseconds was below the specified five-second limit.

The observed availability was 100% during the test period. This short-duration result does not establish long-term production availability.

7. Responsive UI Validation

Responsive UI validation was performed to evaluate the application's behaviour across different screen dimensions.

The following viewport sizes were considered:

Device Category	Resolution
Desktop	1920 × 1080
Laptop	1366 × 768
Tablet Landscape	1024 × 768
Tablet Portrait	768 × 1024
Mobile	390 × 844
Compact Mobile	360 × 800
Narrow Mobile	320 × 568
Mobile Landscape	800 × 360

The frontend automated test suite produced the following results:

Metric	Result
Total frontend tests	11
Passed	11

Failed	0

Browser-based verification was performed on desktop, laptop, tablet landscape, tablet portrait, and mobile landscape layouts.

The first-message interaction path was verified to require no more than three clicks or taps after login. The keyboard interaction path required zero mouse clicks.

Physical-device testing on actual iOS and Android devices remained pending.

8. Overall Validation Summary

Testing Area	Result
Backend automated testing	141 passed
Performance testing	PASS
Concurrent user handling	100 users connected
Message delivery	100%
p95 message latency	70.029 ms
Automatic reconnection	PASS
Message recovery	PASS
Short-duration availability	100%
Frontend automated testing	11 passed
Responsive UI verification	Partially completed

9. Limitations

The following limitations were identified during validation:

- Physical-device testing on actual iOS and Android devices was pending.
- Availability testing covered a short observation period and cannot establish long-term system uptime.
- Performance results represent the recorded 100-user test scenario.
- Automated tests do not replace complete end-to-end acceptance testing of every possible user workflow.

10. Conclusion

The system validation activities demonstrated that the Real-Time Chat Application performed successfully under the tested conditions.

The application supported 100 concurrent simulated users and delivered all 250 expected messages without message loss. The recorded p95 message delivery latency was 70.029 milliseconds, satisfying the specified 200-millisecond requirement.

Reliability testing demonstrated successful automatic reconnection within the required five-second limit, with no message loss during the recorded test period.

The backend and frontend automated test suites also completed successfully. Responsive UI verification was performed across multiple screen dimensions, although physical-device testing remained pending.

Based on the available test results, the application satisfied the evaluated performance and reconnection requirements, with the remaining limitations documented for further testing.